Scrambled states survive local Pauli noise up to the hashing entropy

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Information is encoded in a random pure state from a state 2-design, such as a random stabilizer state, and one layer of Pauli noise then acts without further randomization. We bound the optimal probability of distinguishing two encoded states at every system size by the information spectrum of the Pauli errors. For i.i.d. noise the states stay distinguishable until the error entropy reaches one bit per qubit, with a Gaussian crossover. For local depolarizing noise this threshold, $r = 0.252$, lies well below the $r = 0.423$ that purity bounds certify.

Introduction.—A common model of a quantum computation that scrambles its input is a random state: the encoding is drawn from a state 2-design, an ensemble whose first two moments match Haar-random states, such as random stabilizer states [1]. Hardware noise then acts on the encoded state. How much of the encoded information survives? We ask this in its simplest operational form. Two pure states ψ_1, ψ_2 are drawn independently from a state 2-design. Both pass through the same Pauli channel $\mathcal{N}$, and nothing acts afterward. An observer who knows the two candidates measures the output to decide which one was prepared. With equal priors, the best success probability is $(1 + \overline{T})/2$, where

$$\overline{T} = \mathbb{E}_{\psi_1,\psi_2} \frac{1}{2} \|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1. \tag{1}$$

The order of operations matters. The 2-design describes how the encoded states are chosen; it does not act on the noise. If another 2-design unitary U followed the noise, the effective channel $U^\dagger \mathcal{N}(U \cdot U^\dagger)U$ averaged over U is global depolarization $\rho \mapsto \theta\rho + (1-\theta)I/d$, with $\theta = (d^2 p_I - 1)/(d^2 - 1)$ [2]. For local depolarizing noise $p_I = (1 - 3r/4)^n$, so θ decays exponentially in n for every $r > 0$ and there is no threshold. Without the trailing randomization, the observer can exploit the local structure of the noise. At $n = 8$ and $r = 0.2$, for example, the Haar-averaged distance is 0.736 (Fig. 1), versus about 0.272 after twirling.

This question underlies known costs of quantum error mitigation (QEM). Takagi, Tajima, and Gu [3] and Tsubouchi, Sagawa, and Yoshioka [4] showed that recovering ideal expectation values from noisy samples needs a number of samples that grows exponentially with circuit depth; see also Refs. [5, 6]. Their bounds concern a worst-case pair of inputs and contract a relative entropy toward I/d at every layer, so they predict exponential cost for any noise strength. Quek *et al.* [7] showed that random circuits make this cost appear at very shallow depth. Their mechanism is the one studied here: a 2-design spreads the state over high-weight Pauli operators, which local depolarizing noise of strength r damps by $(1-r)^w$ at weight w. They control the outcome through the purity, which certifies decay only for $(1-r)^2 < 1/3$, i.e., $r > 0.423$.

For the pairwise quantity (1), Ibarrondo and Stilck França [8] proved that the average contraction tends to one when the Choi-state entropy satisfies $S(\tau_\mathcal{N}) < -\log_2 \|\mathcal{N}(I/d_{\text{in}})\|_\infty$; for Pauli channels this means the error entropy is below n bits. Combined with a 2-design upper bound, this gives local-depolarizing thresholds near 0.25 and 0.42, leaving the interval between them open. Liu *et al.* [9] found the same two values by decoupling, with an $O(n^{-1/2})$ upper bound in between. Sommers *et al.* [10] identified the hashing transition and an $n^{-1/2}$ window in the coherent information of random codes.

Here we give two-sided bounds on $\overline{T}$ that hold for every Pauli distribution and system size and are controlled by a single quantity, the information spectrum of the error distribution. For i.i.d. noise they locate the threshold exactly at the hashing entropy, close the interval left open by Refs. [8, 9], and resolve the crossover as a Gaussian profile. The lower bound is the typical-image projector of Ref. [8] applied to Pauli errors; the upper bound splits off the likely errors before taking second moments, as in privacy amplification and quantum soft covering [11–13]. The same splitting explains why purity-based estimates are loose. We also show which conclusions depend only on the design's second moments.

Setup.—Let $d = 2^n$, $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, and

$$\mathcal{N}_p(\rho) = \sum_{P \in \mathcal{P}_n} p_P P\rho P, \qquad s(P) = -\log_2 p_P, \tag{2}$$

where the surprisal s is a random variable with $P \sim p$. The distribution p is arbitrary, so errors may be correlated across qubits. Logarithms are base two. The channel does not depend on the state samples. Each $\psi_i = |\psi_i\rangle\langle\psi_i|$ is drawn independently from an exact pure-state 2-design on the full d-dimensional space:

$$\mathbb{E}\psi_i = I/d, \qquad \mathbb{E}\psi_i^{\otimes 2} = (I + F)/[d(d+1)], \tag{3}$$

where F swaps the two copies. The two ensembles need not coincide.

Theorem 1. *For every Pauli distribution p and $a \geq 0$,*

$$\mathbb{P}[s < n - a] - 2^{-a} \leq \overline{T} \leq \mathbb{P}[s < n + a] + 2^{-a/2}/\sqrt{2}. \tag{4}$$

More generally, for any subset $\mathcal{G} \subseteq \mathcal{P}_n$ and any partition

$\{\mathcal{C}_j\}$ of $\mathcal{P}_n$, with $m_j = \sum_{\mathcal{C}_j} p_P$ and $v_j = \sum_{\mathcal{C}_j} p_P^2$,

$$\overline{T} \geq p(\mathcal{G}) - |\mathcal{G}|/d, \tag{5}$$

$$\overline{T} \leq \sum_j \min\left\{ m_j, \sqrt{\frac{d^2 v_j - m_j^2}{2(d+1)}} \right\}. \tag{6}$$

Proof. Fix ψ_1 and let Π project onto the span of $\{P|\psi_1\rangle : P \in \mathcal{G}\}$. For each realization, the variational formula gives $\frac{1}{2}\|\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)\|_1 \geq \operatorname{tr}\Pi[\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)]$. The first term is at least $p(\mathcal{G})$. Averaging the second over the independent ψ_2 uses only $\mathbb{E}\psi_2 = I/d$ and unitality, giving $\operatorname{tr}\Pi/d \leq |\mathcal{G}|/d$. Averaging over ψ_1 proves (5); only the second state needs to be a 1-design. This is the typical-Kraus-image argument of Ref. [8] (Theorem 18 there), with an arbitrary set of Pauli errors in place of a typical set.

For the upper bound let $\Delta = \psi_1 - \psi_2$ and $\mathcal{N}_j(\rho) = \sum_{\mathcal{C}_j} p_P P\rho P$. Pauli orthogonality and (3) give

$$\mathbb{E}\operatorname{tr}\mathcal{N}_j(\psi_i)^2 = \frac{dv_j + m_j^2}{d+1}, \qquad \mathbb{E}\operatorname{tr}\mathcal{N}_j(\psi_1)\mathcal{N}_j(\psi_2) = \frac{m_j^2}{d},$$
$$\mathbb{E}\operatorname{tr}\mathcal{N}_j(\Delta)^2 = \frac{2d}{d+1}v_j - \frac{2m_j^2}{d(d+1)}. \tag{7}$$

Jensen's inequality and $\|A\|_1 \leq \sqrt{d}\|A\|_2$ give the square-root term in (6); the triangle inequality gives $\frac{1}{2}\|\mathcal{N}_j(\Delta)\|_1 \leq m_j$. Sum over classes.

For (4), take $\mathcal{G} = \{s < n - a\}$, which has fewer than $d2^{-a}$ elements, and split the partition at $s = n+a$: bound the head by its mass and the tail by $\sqrt{dv/2}$ with $v \leq 2^{-n-a}$. □

The two cumulative probabilities in (4) are evaluated at $n \mp a$, so the bracket is tight whenever the surprisal concentrates. The event $s < n$ means $p_P > 1/d$: the likely errors are those whose images $P|\psi_1\rangle$ fit into the d-dimensional space without overlapping. The lower bound is positive only when some error has $p_P > 1/d$, which for i.i.d. noise requires $\max_i p_i > 1/2$, i.e., $r < 2/3$ for depolarizing noise.

The single-class partition, $\mathcal{C} = \mathcal{P}_n$, reduces (6) to the purity of the noisy output, $\overline{T} \leq [d\sum_P p_P^2/2]^{1/2}$. This is the quantity that controls the error-mitigation bounds of Ref. [7]. For local depolarizing noise $\sum_P p_P^2 = [(1 + 3(1-r)^2)/4]^n$, which decays faster than d^{-1} only for $r > 1 - 1/\sqrt{3} = 0.423$. Splitting off the likely errors is what lets the bracket follow the distance below that value.

Hashing threshold for i.i.d. noise.—Let $p = \boldsymbol{p}^{\otimes n}$ with single-qubit distribution $\boldsymbol{p} = (p_0, p_x, p_y, p_z)$, entropy $H = \mathbb{E}[-\log_2 p_i]$, and variance $\sigma^2 = \operatorname{Var}[-\log_2 p_i]$. The surprisal is then a sum of n i.i.d. terms with mean nH. For $H < 1$, choosing $a = \epsilon n$ with $0 < \epsilon < 1 - H$ and a large-deviation bound [14] give $\overline{T} \geq 1 - e^{-\Omega(n)}$. For $H > 1$ the upper bound gives $\overline{T} \leq e^{-\Omega(n)}$. The threshold $H = 1$ is the zero of the coherent information $1 - H$ of the Pauli channel at maximally mixed input [15, 16], the hashing point. (Degenerate codes can transmit quantum information slightly beyond it [17]; the present threshold concerns discrimination of two encoded states.) For depolarizing noise $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, with $q = 3r/4$ and $\boldsymbol{p} = (1-q, q/3, q/3, q/3)$, the threshold is $q_c = 0.189290\ldots$, or $r_c = 0.252386\ldots$. Local depolarizing noise is therefore tolerated up to r_c, whereas the purity bound above certifies loss only beyond 0.423.

Exponential decay above r_c can also be obtained from decoupling. The unsmoothed Rényi bound of Ref. [12] applies because, for any reference-system operators X_P, Eq. (3) gives

$$\mathbb{E}_\psi \Big\| \sum_P (P\psi P - I/d) \otimes X_P \Big\|_2^2 = \frac{d}{d+1}\sum_P \|X_P\|_2^2 - \frac{\|\sum_P X_P\|_2^2}{d(d+1)} \leq \|X\|_2^2. \tag{8}$$

Theorem 8 of Ref. [12] then gives, for $1 < \alpha \leq 2$,

$$\mathbb{E}\|\mathcal{N}_p(\psi) - I/d\|_1 \leq 2^{2/\alpha - 1} 2^{-\frac{\alpha-1}{\alpha}[H_\alpha(p) - n]}, \tag{9}$$

which decays once α is close enough to one that $H_\alpha(\boldsymbol{p}) > 1$. The smoothed bound of Ref. [9] gives the same with smoothing parameter $2^{-\kappa n}$ for small $\kappa > 0$, via $H_{\min} \leq H_2$ [Eq. (48) of Ref. [18]] and the AEP [19]. Decoupling gives the decay but not the location of the transition or its shape; the bracket gives both.

Let the single-qubit distribution $\boldsymbol{p}_n$ depend on n, with i.i.d. errors at each n, and set $x_n = \sqrt{n}(1 - H_n)/\sigma_n$.

Corollary 1. *If $\boldsymbol{p}_n \to \boldsymbol{p}_*$ with $H(\boldsymbol{p}_*) = 1$ and $\sigma(\boldsymbol{p}_*) > 0$, then uniformly over independent exact 2-design inputs,*

$$\overline{T}_n = \Phi(x_n) + O\left(\frac{\log n}{\sqrt{n}}\right), \tag{10}$$

where Φ is the standard normal distribution function.

Proof. With $\rho_n = \mathbb{E}|-\log_2 p_{n,i} - H_n|^3$ and $b_n = C_{\mathrm{BE}}\rho_n/(\sigma_n^3\sqrt{n})$, the Berry–Esseen theorem [20] applied to (4) gives

$$\overline{T}_n \geq \Phi\left(x_n - \frac{a}{\sqrt{n}\sigma_n}\right) - b_n - 2^{-a},$$
$$\overline{T}_n \leq \Phi\left(x_n + \frac{a}{\sqrt{n}\sigma_n}\right) + b_n + 2^{-a/2}/\sqrt{2}. \tag{11}$$

On a four-letter alphabet the third moment is bounded uniformly, since $t|\log t|^3 \to 0$ at zero, and σ_n stays bounded away from zero. Hence $b_n = O(n^{-1/2})$; set $a = 2\log_2 n$. □

A transverse crossing of $H = 1$ therefore has a noise window of width $n^{-1/2}$ with the universal profile Φ. Figure 1 shows the finite-size bracket and the approach to Φ.

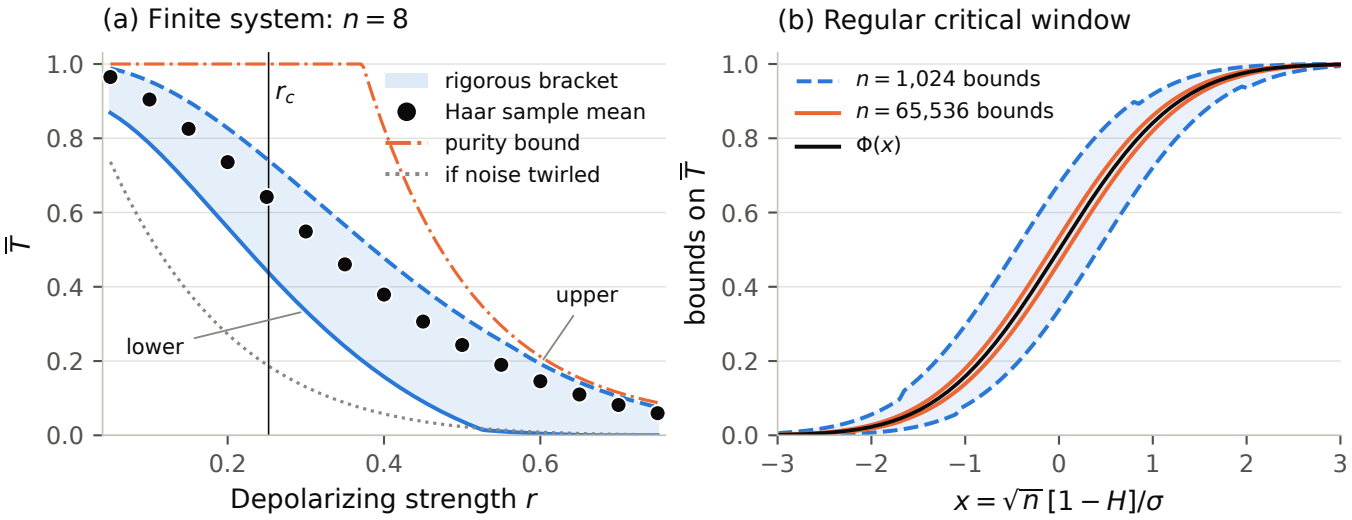


FIG. 1. Distinguishability after one layer of local depolarizing noise. (a) At $n = 8$, the shaded interval lies between the projector lower bound (solid) and the two-class upper bound (dashed) of Eqs. (5)–(6), optimized over error-weight cutoffs; the thin vertical line marks r_c. The lower bound vanishes for $r \geq 2/3$, where no single error has probability above $1/d$. Points are means over 64 independent Haar-state pairs, computed from full density matrices; bars are one sample standard error, and the same pairs are reused at every r. The dash-dotted curve is the single-class (purity) bound, which falls below one only for $r \gtrsim 0.37$ at this size and certifies decay only above 0.423 asymptotically. The dotted gray curve is the depolarizing parameter θ that would result if a further 2-design unitary twirled the noise; it equals the twirled distance up to a factor $1 - O(1/d)$. (b) The bracket (4) with $a = 2\log_2 n$, evaluated from binomial distributions at fixed $x = \sqrt{n}(1 - H)/\sigma$, for $n = 1024$ and 65536. The intervals approach $\Phi(x)$ (black). Panel (b) shows bounds, not simulations.

Correlated noise.—For correlated errors the bracket still applies, and the relevant quantity is the full information spectrum rather than the entropy. Global depolarization $\mathcal{N}(\rho) = \theta\rho + (1-\theta)I/d$, $0 < \theta < 1$, has $p_I = \theta + (1-\theta)/d^2$ and $p_P = (1-\theta)/d^2$ otherwise. Its entropy rate tends to $2(1-\theta)$, above one for $\theta < 1/2$, yet $\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2) = \theta(\psi_1 - \psi_2)$ gives $\overline{T} \to \theta$. The bracket recovers this limit: with $a = n/2 - 1$, only the identity has surprisal below $n - a$, every other error lies above $n + a$, and $|\overline{T} - \theta| \leq 2^{-n/4+1}$ for large n.

Design dependence.—The bounds above hold for every exact 2-design. Two finer features depend on higher moments of the ensemble, which sets the reach of statements made for an arbitrary design of a given order [8, 9].

Degenerate critical noise. The variance condition in Corollary 1 cannot be dropped. Complete dephasing, $\boldsymbol{p}_* = (1/2, 0, 0, 1/2)$, has $H = 1$ and $\sigma = 0$. In dimension $d = 2^n$ the uniform union of $d + 1$ mutually unbiased bases (MUBs) is an exact 2-design [21]; take the computational basis as one of them. Its d states are unchanged by dephasing, and every other state maps to I/d. Counting pairs,

$$\overline{T}_{\mathrm{MUB}} = \frac{(d-1)(2d+1)}{d(d+1)^2} \longrightarrow 0. \tag{12}$$

For Haar states the dephased outputs are diagonal, so their trace distance is the total-variation distance between two independent Dirichlet$(1, \ldots, 1)$ vectors. Single coordinates $X, Y \sim \mathrm{Beta}(1, d-1)$ satisfy $\mathbb{E}\min(X, Y) = \int_0^1 (1-t)^{2d-2} dt = (2d-1)^{-1}$, hence

$$\overline{T}_{\mathrm{Haar}} = \frac{d}{2}\mathbb{E}|X - Y| = \frac{d-1}{2d-1} \longrightarrow \frac{1}{2}. \tag{13}$$

The two ensembles share their first two moments, yet their limits differ (Fig. 2). This strengthens the observation of Ref. [8] that computational-basis states, a 1-design, are not contracted by dephasing. Positive variance at each n is also not enough. Set $\epsilon_n = n^{-6}$ and $\boldsymbol{p}_n = (1/2 - \epsilon_n + \delta_n, \epsilon_n, \epsilon_n, 1/2 - \epsilon_n - \delta_n)$, with $\delta_n > 0$ solving $H(\boldsymbol{p}_n) = 1$. Binary entropy bounds give $\delta_n = O(\sqrt{\epsilon_n \log(1/\epsilon_n)})$, so $n\|\boldsymbol{p}_n - \boldsymbol{p}_*\|_1 \to 0$. Replacing one tensor factor at a time and using $\|\mathcal{A}\otimes\mathcal{B} - \mathcal{A}'\otimes\mathcal{B}'\|_\diamond \leq \|\mathcal{A} - \mathcal{A}'\|_\diamond + \|\mathcal{B} - \mathcal{B}'\|_\diamond$, the n-qubit channels differ in diamond norm by at most this quantity. MUB inputs still give $\overline{T}_n \to 0$, although $\sigma_n > 0$ and $x_n = 0$ for every n.

Decay rates. For i.i.d. errors, grouping Pauli strings into empirical types t in (6) gives $\overline{T}_n \leq \mathrm{poly}(n) 2^{-nE(\boldsymbol{p})}$, using standard type bounds [14], with

$$\begin{aligned} E(\boldsymbol{p}) &= \min_{t \ll \boldsymbol{p}} \{D(t\|\boldsymbol{p}) + \tfrac{1}{2}[H(t) - 1]_+\} \\ &= \max_{0 \leq s \leq 1} \frac{s}{1+s}[H_{1+s}(\boldsymbol{p}) - 1]. \end{aligned} \tag{14}$$

Here $D(t\|p) = \sum_i t_i \log_2(t_i/p_i)$ and $H_\alpha(p) = (1-\alpha)^{-1}\log_2 \sum_i p_i^\alpha$. The second line follows by writing $[H-1]_+/2 = \max_{0 \leq u \leq 1/2} u(H-1)$, exchanging extrema,

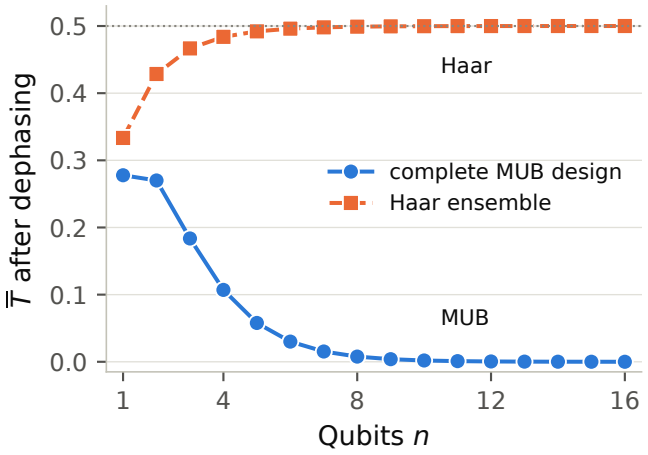


FIG. 2. Design dependence at a degenerate critical channel. Under complete dephasing both ensembles are exact 2-designs and the error entropy is one bit per qubit. Equations (12) and (13), evaluated without sampling, give different limits. The dotted gray line marks 1/2.

and setting $s = u/(1-u)$. This is the known universal$_2$ privacy-amplification exponent at rate one [11], and Eq. (9) reproduces it without the polynomial prefactor. Applying (5) to empirical types t with $D(t\|\boldsymbol{p})+H(t) < 1$ gives a matching lower bound near threshold. With $R_{\mp}$ the lim inf and lim sup of $-n^{-1}\log_2 \overline{T}_n$,

$$E(\boldsymbol{p}) \le R_- \le R_+ \le \inf_{D(t\|\boldsymbol{p})+H(t)<1} D(t\|\boldsymbol{p}). \tag{15}$$

For depolarizing noise both sides expand to the same leading order: $R_\pm = \kappa(r-r_c)^2 + O((r-r_c)^3)$ with $\kappa = 9/[32\ln 2\, q_c(1-q_c)] = 2.64\ldots$. Away from threshold the rate depends on the ensemble. Uniformly random stabilizer states can be written as $\psi = d^{-1}[I + \sum_{P\neq I} z_P P]$ with $z_P \in \{0, \pm 1\}$ and $\mathbb{E}|z_P| = 1/(d+1)$. Local depolarization multiplies a weight-w term by $(1-r)^w$, and the triangle inequality gives

$$\overline{T}_n^{\,\mathrm{stab}} \le \frac{[1+3(1-r)]^n - 1}{2^n + 1}. \tag{16}$$

At $r = 0.9$ this rate is at least $1 - \log_2(1.3) = 0.621\ldots$, larger than $E = 0.479\ldots$ from (14). Random stabilizer states form a 3-design [1], so even three matching moments leave the decay rate ensemble-specific; determining the Haar rate away from threshold is an open problem.

Discussion.—Scrambling an encoding does not make it fragile to local Pauli noise: the information survives up to the hashing entropy, where the error distribution carries one bit per qubit. A single bracket, built from the error distribution and the dimension alone, captures this for every exact 2-design and for correlated noise, and gives the transition its Gaussian shape. For error mitigation, Refs. [3, 4, 7] show exponential cost at large depth for any noise strength. After one noisy layer, the purity-based estimate behind such bounds is loose throughout $0.252 < r < 0.423$, because it treats the likely low-surprisal errors, which the optimal measurement exploits, in the same way as the unlikely ones. Splitting the error distribution at surprisal n, as done here, is a natural way to sharpen those estimates. Natural next steps are worst-case versions relevant to error-mitigation cost, approximate designs from shallow circuits, and non-Pauli noise.

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Data availability.—The data that support the findings of this article and the code that generates every figure and numerical check accompany this submission and will be deposited in a public repository [22].

[1] R. Kueng and D. Gross, Qubit stabilizer states are complex projective 3-designs, arXiv:1510.02767 (2015).
[2] C. Dankert, R. Cleve, J. Emerson, and E. Livine, Exact and approximate unitary 2-designs and their application to fidelity estimation, Phys. Rev. A **80**, 012304 (2009).
[3] R. Takagi, H. Tajima, and M. Gu, Universal sampling lower bounds for quantum error mitigation, Phys. Rev. Lett. **131**, 210602 (2023).
[4] K. Tsubouchi, T. Sagawa, and N. Yoshioka, Universal cost bound of quantum error mitigation based on quantum estimation theory, Phys. Rev. Lett. **131**, 210601 (2023).
[5] R. Takagi, Optimal resource cost for error mitigation, Phys. Rev. Res. **3**, 033178 (2021).
[6] R. Takagi, S. Endo, S. Minagawa, and M. Gu, Fundamental limits of quantum error mitigation, npj Quantum Inf. **8**, 114 (2022).
[7] Y. Quek, D. Stilck França, S. Khatri, J. J. Meyer, and J. Eisert, Exponentially tighter bounds on limitations of quantum error mitigation, Nat. Phys. **20**, 1648 (2024).
[8] R. Ibarrondo and D. Stilck França, Average contraction coefficients of quantum channels, arXiv:2508.08214 (2025).
[9] G. Liu, C. Qin, Z. Xu, X. Ma, and Z.-W. Liu, Can scrambling protect quantum state distinguishability under noise?, arXiv:2606.02122 (2026).
[10] G. M. Sommers, J. A. Jacoby, Z. Weinstein, D. A. Huse, and S. Gopalakrishnan, Spectral properties and coding transitions of Haar-random quantum codes, PRX Quan-

tum **7**, 020328 (2026).
[11] M. Hayashi, Tight exponential analysis of universally composable privacy amplification and its applications, IEEE Trans. Inf. Theory **59**, 7728 (2013).
[12] F. Dupuis, Privacy amplification and decoupling without smoothing, IEEE Trans. Inf. Theory **69**, 7784 (2023).
[13] Y.-C. Shen, L. Gao, and H.-C. Cheng, Optimal second-order rates for quantum soft covering and privacy amplification, IEEE Trans. Inf. Theory **70**, 5077 (2024).
[14] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, Hoboken, 2006), Ch. 11.
[15] C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, and W. K. Wootters, Mixed-state entanglement and quantum error correction, Phys. Rev. A **54**, 3824 (1996).
[16] I. Devetak, The private classical capacity and quantum capacity of a quantum channel, IEEE Trans. Inf. Theory **51**, 44 (2005).
[17] G. Smith and J. A. Smolin, Degenerate quantum codes for Pauli channels, Phys. Rev. Lett. **98**, 030501 (2007).
[18] M. Tomamichel, M. Berta, and M. Hayashi, Relating different quantum generalizations of the conditional Rényi entropy, J. Math. Phys. **55**, 082206 (2014).
[19] M. Tomamichel, R. Colbeck, and R. Renner, A fully quantum asymptotic equipartition property, IEEE Trans. Inf. Theory **55**, 5840 (2009).
[20] W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. II, 2nd ed. (Wiley, New York, 1971), Sec. XVI.5.
[21] A. Klappenecker and M. Rötteler, Mutually unbiased bases are complex projective 2-designs, in *Proceedings of the 2005 IEEE International Symposium on Information Theory* (IEEE, 2005), pp. 1740–1744.
[22] X. Zhao, Data and code for "Scrambled states survive local Pauli noise up to the hashing entropy" [repository and DOI to be assigned on deposit].